

Pair of Linear Equations in Two Variables

How to tell whether two linear equations have one solution, no solution or infinitely many, how to solve them by graphs and by algebra, and how to use them to solve word problems, with worked examples and practice questions.

Introduction

A **linear equation in two variables** is an equation of the form $ax + by + c = 0$, where a , b and c are real numbers and a and b are not both zero. For example, $2x + 3y = 7$ and $x - 2y - 4 = 0$ are linear equations in x and y . Each such equation has infinitely many solutions, and its graph is a straight line.

When we have **two** such equations together, using the same variables, we get a **pair of linear equations in two variables**:

$$a_1x + b_1y + c_1 = 0 \quad \text{and} \quad a_2x + b_2y + c_2 = 0$$

A solution of the pair is a pair of values (x, y) that satisfies both equations at the same time. Geometrically, it is the point where the two lines meet, if they meet at all. In this chapter we learn how to find that point, or to explain why no such point exists, or why there are infinitely many.

Graphical Method

Since each equation in the pair represents a straight line, we can draw both lines on the same graph. Each line is drawn by plotting two or three points that satisfy that equation. There are only three possibilities:

- The two lines **intersect** at one point, there is exactly one solution.
- The two lines are **parallel**, they never meet, so there is no solution.

- The two lines **coincide** (lie on top of each other), every point on the line satisfies both equations, so there are infinitely many solutions.

In the classroom, the graphical method is a good way to see which case you are in. In an exam, drawing the graph neatly with at least two points per line is expected if the question asks for it, but for accurate answers algebraic methods are usually faster.

For example, the pair $x + y = 5$ and $2x + 2y = 10$ represents the same line (the second equation is just twice the first), so the lines coincide. The pair $x + y = 5$ and $x + y = 7$ represents two parallel lines, no point satisfies both. And the pair $x + y = 5$ and $x - y = 1$ gives two lines that cross at $(3, 2)$, which is the unique solution.

Consistency and Number of Solutions

Before solving, we can tell which of the three cases we are in just by comparing the coefficients. This is one of the most frequently tested ideas in the chapter.

Conditions for solutions, given $a_1x + b_1y + c_1 = 0$ and $a_2x + b_2y + c_2 = 0$

Case 1, Intersecting lines (unique solution):

$$a_1/a_2 \neq b_1/b_2$$

The pair is **consistent** and has exactly one solution.

Case 2, Parallel lines (no solution):

$$a_1/a_2 = b_1/b_2 \neq c_1/c_2$$

The pair is **inconsistent** and has no solution.

Case 3, Coincident lines (infinitely many solutions):

$$a_1/a_2 = b_1/b_2 = c_1/c_2$$

The pair is **consistent (dependent)** and has infinitely many solutions.

A common mistake is to forget the c terms. You need all three ratios to distinguish Case 2 from Case 3. If $a_1/a_2 = b_1/b_2$, you must then check whether that common ratio equals c_1/c_2 or not.

The words to remember: **consistent** means at least one solution exists (Cases 1 and 3); **inconsistent** means no solution (Case 2).

Substitution Method

The substitution method is exactly what it sounds like: solve one equation for one variable and substitute that expression into the other equation.

Substitution method, steps

Step 1. From one equation, express one variable in terms of the other (usually choose the variable with coefficient 1 or -1 to keep the algebra simple).

Step 2. Substitute this expression into the other equation. You now have one equation in one variable, solve it.

Step 3. Put the value found in Step 2 back into the expression from Step 1 to get the other variable.

Worked Example 1, Substitution method

Solve $x + 2y = 6$ and $3x + 4y = 14$ by substitution.

From the first equation: $x = 6 - 2y$.

Substitute into the second equation:

$$3(6 - 2y) + 4y = 14 \implies 18 - 6y + 4y = 14 \implies -2y = -4 \implies y = 2$$

Put $y = 2$ back into $x = 6 - 2y$:

$$x = 6 - 2(2) = 2$$

So the solution is **$x = 2, y = 2$** .

Check: $3(2) + 4(2) = 6 + 8 = 14 \checkmark$ and $2 + 2(2) = 6 \checkmark$.

Elimination Method

In the elimination method, we make the coefficients of one variable the same (or opposites) in both equations, then add or subtract to remove that variable entirely.

Elimination method, steps

Step 1. Multiply one or both equations by a suitable number so that the coefficients of one variable become equal (or equal and opposite).

Step 2. Add the equations if the coefficients are opposite in sign, or subtract if they are the same sign, to eliminate one variable.

Step 3. Solve the resulting equation for the remaining variable, then substitute back to find the other.

Elimination is often the quickest method when both equations are already in standard form and neither variable has coefficient 1.

Worked Example 2, Elimination method

Solve $2x + 3y = 11$ and $2x - 4y = -24$ by elimination.

Both equations have $2x$, so subtract the second from the first to eliminate x :

$$(2x + 3y) - (2x - 4y) = 11 - (-24) \implies 7y = 35 \implies y = 5$$

Substitute $y = 5$ into the first equation:

$$2x + 3(5) = 11 \implies 2x = 11 - 15 = -4 \implies x = -2$$

So the solution is **$x = -2$, $y = 5$** .

Check: $2(-2) - 4(5) = -4 - 20 = -24 \checkmark$.

Word Problems

Many real-life situations reduce to a pair of linear equations, a cost problem, an age problem, a speed-distance problem, or a fraction problem. The steps are always the same: choose variables, translate the words into two equations, solve the pair by any method, and check that the answer makes sense in the original context.

Worked Example 3, Word problem

The sum of the ages of a father and his son is 45 years. Five years ago, the father was 4 times as old as his son. Find their present ages.

Let the father's present age be x years and the son's present age be y years.

From "the sum is 45":

$$x + y = 45 \quad \dots(1)$$

Five years ago, the father was $(x - 5)$ and the son was $(y - 5)$. The condition gives:

$$x - 5 = 4(y - 5) \implies x - 5 = 4y - 20 \implies x - 4y = -15 \quad \dots(2)$$

Subtract (2) from (1):

$$(x + y) - (x - 4y) = 45 - (-15) \implies 5y = 60 \implies y = 12$$

Put $y = 12$ into (1): $x = 45 - 12 = 33$.

So the **father is 33 years old and the son is 12 years old**.

Check: five years ago, father was 28 and son was 7, and $28 = 4 \times 7 \checkmark$.

Summary

- A pair $a_1x + b_1y + c_1 = 0$, $a_2x + b_2y + c_2 = 0$ represents two straight lines.
- Graphically: intersecting lines $\rightarrow$ one solution; parallel lines $\rightarrow$ no solution; coincident lines $\rightarrow$ infinitely many solutions.
- **Consistency conditions:** $a_1/a_2 \neq b_1/b_2 \rightarrow$ unique solution (consistent); $a_1/a_2 = b_1/b_2 \neq c_1/c_2 \rightarrow$ no solution (inconsistent); $a_1/a_2 = b_1/b_2 = c_1/c_2 \rightarrow$ infinitely many solutions (consistent, dependent).
- **Substitution:** express one variable from one equation and substitute into the other.
- **Elimination:** make coefficients of one variable equal and add or subtract to eliminate it.
- Word problems: define variables, form two equations from the given conditions, solve the pair, and verify in the original context.

Practice Questions

1. Check whether the pair $x + 2y = 3$ and $2x + 4y = 8$ is consistent. What does the result tell you about the two lines?

2. Solve $3x + y = 5$ and $2x - y = 5$ by the substitution method.

3. Solve $x + y = 7$ and $3x - 2y = 11$ by the elimination method.

4. Solve $2x + 3y = 7$ and $4x + 6y = 14$. How many solutions does this pair have, and why?

5. Five years so, the age of a mother will be three times the age of her daughter. Five years ago, the mother's age was seven times the daughter's age. Find their present ages.